GRADE 7

Computational Thinking

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **CHECK YOUR NAME, CLASS, ROLL NUMBER AND OTHER DETAILS ON THE OMR ANSWER SHEET.**
- **USE ONLY BLACK/BLUE BALL PEN AND HB PENCIL TO FILL THE OMR ANSWER SHEET.**
- **ALL 50 QUESTIONS ARE COMPULSORY AND CARRY ONE MARK EACH.**
- **SECTION A: ABSTRACT & SPATIAL THINKING (10); SECTION B: PATTERN RECOGNITION (10); SECTION C: DECOMPOSITION, NUMBERS & DATA (10); SECTION D: ALGORITHMIC THINKING & LOGICAL SEQUENCING (10); SECTION E: OLYMPIAD CHALLENGE (10).**
- **ONLY OMR ANSWER SHEETS WILL BE EVALUATED. THERE IS NO NEGATIVE MARKING.**
- **CALCULATORS, SMART WATCHES AND MOBILE PHONES ARE NOT ALLOWED.**
- **USE AVAILABLE MARGINS/SPACE FOR ROUGH WORK. SUBMIT THE OMR SHEET AND QUESTION PAPER AFTER THE EXAM.**

NAME	ROLL NO.	CLASS / SECTION	SIGNATURE

Sample Paper

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. A robot starts facing North. It moves 5 squares, turns right and moves 3, turns right and moves 2, then turns left and moves 4. Where is it from its starting point?

A. North-East
B. North-West
C. South-East
D. South-West

Q2. Four identical cubes are arranged as a 2 × 2 square block in one layer. How many cube faces are exposed altogether?

A. 14
B. 16
C. 18
D. 20

Q3. A figure is reflected first in a vertical mirror and then in a horizontal mirror through the same centre. The combined effect equals a rotation of:

A. 45°
B. 90°
C. 180°
D. 270°

Q4. On a cube, Red is opposite Blue, Green is opposite Yellow, and Black is opposite White. If Red is on top and Green is in front, which colour cannot be on the right face?

A. Blue
B. Black
C. White
D. Either Black or White

Q5. A sheet is folded in half three times. One hole is punched away from all folds. How many holes appear when it is fully opened?

A. 4
B. 6
C. 8
D. 12

Q6. An equilateral triangle is rotated about its centre. What is the smallest positive angle that makes it look unchanged?

A. 60°
B. 90°
C. 120°
D. 180°

Q7. A regular hexagon has how many lines of symmetry?

A. 3
B. 4
C. 6
D. 8

Q8. A cuboid has 6 faces and 8 vertices. How many edges does it have?

A. 10
B. 12
C. 14
D. 16

Q9. An arrow points South-East. It is rotated 225° clockwise. Which direction will it point?

A. North
B. North-East
C. West
D. South-West

Q10. On a map, 2 cm represents 5 km. Two places are 7.2 cm apart on the map. What is the actual distance?

A. 16 km
B. 18 km
C. 20 km
D. 22 km

Sample paper

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 4, 9, 19, 39, 79, ___.

A. 119
B. 139
C. 159
D. 169

Q12. Find the next number: 2, 6, 15, 31, 56, ___.

A. 82
B. 88
C. 92
D. 96

Q13. Which letter comes next? A, D, H, M, S, ___.

A. X
B. Y
C. Z
D. A

Q14. A number rule gives 2 → 9, 3 → 16, 4 → 25. Using the same rule, 6 → ?

A. 36
B. 42
C. 49
D. 56

Q15. Find the next number: 1, 2, 6, 24, 120, ___.

A. 360
B. 480
C. 600
D. 720

Q16. Find the next number: 100, 50, 25, 12.5, 6.25, ___.

A. 2.125
B. 3.125
C. 3.25
D. 4.125

Q17. Find the next number: 2, 3, 5, 8, 12, 17, ___.
A. 21
B. 22
C. 23
D. 24

Q18. Find the next pair: A1, C4, F9, J16, O25, ___.
A. T30
B. U36
C. V36
D. U49

Q19. Find the next number: 81, 27, 9, 3, 1, ___.
A. 1/2
B. 1/3
C. 1/4
D. 0

Q20. Find the next number: 6, 18, 16, 48, 46, 138, 136, ___.
A. 272
B. 386
C. 408
D. 410

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q21. The ratio of boys to girls in a club is 3 : 5. If there are 64 students altogether, how many are boys?

A. 20
B. 24
C. 32
D. 40

Q22. Forty percent of 250 cards are red. If 30 red cards are removed, how many red cards remain?

A. 60
B. 70
C. 80
D. 90

Q23. What is the smallest positive number divisible by 6, 8 and 10?

A. 60
B. 80
C. 120
D. 240

Q24. Four daily readings are 18, 24, 30 and 36. What fifth reading makes the average of all five readings equal to 28?

A. 28
B. 30
C. 32
D. 34

Q25. A tank is 3/5 full and contains 48 L. If 12 L is removed, what fraction of the whole tank remains filled?

A. 3/10
B. 2/5
C. 9/20
D. 1/2

Q26. A bus leaves at 8:35 a.m., travels 1 h 47 min, stops for 28 min, then travels 56 min. When does it arrive?

A. 11:36 a.m.
B. 11:46 a.m.
C. 11:56 a.m.
D. 12:06 p.m.

Q27. Three-eighths of 120 students joined a coding club. How many students joined?

A. 40
B. 45
C. 48
D. 50

Q28. A table records 24, 31, 28 and 37 books read by four houses. How many books were read altogether?

A. 110
B. 115
C. 120
D. 125

Q29. Two groups are in the ratio 4 : 5 and contain 72 students altogether. How many students are in the larger group?

A. 32
B. 36
C. 40
D. 45

Q30. Two-thirds of a number is 42. What is the number?

A. 56
B. 60
C. 63
D. 72

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q31. A number machine multiplies the input by 3, subtracts 4, then divides by 2. What is the output for input 10?
A. 11
B. 12
C. 13
D. 14

Q32. A machine doubles a number, adds 5 and then subtracts 3. The final output is 30. What was the input?
A. 12
B. 13
C. 14
D. 15

Q33. Use this rule four times: if the number is even, divide by 2; if it is odd, multiply by 3 and add 1. Start with 7. What is the final number?
A. 11
B. 14
C. 17
D. 22

Q34. A robot moves 4 squares East, turns left and moves 3, turns left and moves 1, then turns right and moves 2. Where is it from the start?
A. North-East
B. North-West
C. South-East
D. South-West

Sample paper

Q35. Four tasks follow these rules: S is before P; P is immediately before Q; R is after Q. Which task must be third?
A. P
B. Q
C. R
D. S

Q36. A robot should draw a square by repeating [Move 5 steps, Turn Right] four times. The program repeats the block only three times. What is the best correction?
A. Repeat it 2 times
B. Repeat it 4 times
C. Change Right to Left
D. Add one Move command only

Q37. Start with 10. If even, divide by 2; if odd, add 7. Apply the rule four times. What is the final number?
A. 3
B. 5
C. 6
D. 12

Q38. Set Total = 0. For n = 1 to 6, add n if n is odd; otherwise add 2n. What is Total?
A. 27
B. 30
C. 33
D. 36

Q39. A robot must make exactly 3 moves Right and 2 moves Up. How many different shortest routes are possible?
A. 6
B. 8
C. 10
D. 12

Q40. A = 5 and B = 8. Execute C = A; A = B; B = C. What are the final values?
A. A=5, B=8
B. A=8, B=5
C. A=8, B=8
D. A=5, B=5

SECTION E – OLYMPIAD CHALLENGE

Q41. An AI system predicts tomorrow's temperature from past numerical weather data. Which technique is mainly being used?
A. Classification
B. Regression
C. Clustering
D. Encryption

Q42. An AI groups customers with similar buying habits without being given group labels. Which technique is this?
A. Classification
B. Regression
C. Clustering
D. Sorting

Q43. Which AI domain is most directly used by an automatic language-translation tool?
A. Computer Vision
B. Natural Language Processing
C. Robotics
D. Data Storage

Q44. A hiring AI is trained mostly on examples from one group and performs poorly for others. What is the main concern?
A. Compression
B. Bias and fairness
C. Faster processing
D. File size

Q45. I am a 3-digit number. My hundreds digit is 2 more than my tens digit; my units digit is twice my tens digit; my digits add to 14. What number am I?
A. 425
B. 536
C. 647
D. 758

Q46. A robot must make exactly 3 moves Right and 2 moves Up. How many different shortest routes can it take?
A. 6
B. 8
C. 10
D. 12

Q47. Which is fixed-rule automation rather than AI?
A. Voice assistant understanding speech
B. Recommendation system learning preferences
C. Light switching on daily at 6:30 p.m.
D. Object recognition in photos

Q48. Which is an example of supervised learning?
A. Grouping unlabeled songs
B. Learning from images labelled cat/dog
C. Following a timer
D. Sorting files by size

Q49. Which practice best protects online privacy?
A. Reusing one easy password
B. Sharing passwords with friends
C. Using strong unique passwords
D. Posting personal details publicly

Q50. A recorded voice note is primarily what type of data?
A. Text
B. Image
C. Sound
D. Numerical

Sample paper

Sample paper